

Maaz Malik
CSE (AIML)

Company: IDFC First Bank

Job Profile: Application Engineer

Offer Type: [Internship + Placement]

Internship Stipend: unspecified (rumoured to be 40k)

CTC: 18.10 LPA (Fixed: 14 LPA + Joining Bonus: 2 L + Variable: 2.10 LPA)

Location: Mumbai/Bangalore/Hyderabad

Process Date: PPT(25th July, 2025) + OA(28th July, 2025) + Interviews(30th July, 2025)

Placement Process:

1. OA Round
2. Technical Round 1
3. Technical Round 2
4. HR Round
5. Technical Round 3

1. Assessment Round (140 Candidates)

This was conducted on the HirePro platform.

1. general aptitude / reasoning - 20 min
2. Core cs fundamentals - 30 min
3. Coding Section - 40 min

Core cs fundamentals involved mcqs based on code snippets/flowcharts, DBMS, OS, SQL. It was

completely language independent and basic level questions were asked.

DSA

Q1) How many elements are correctly sorted.

Correct place means Rightly indexed when seen from a Sorted perspective.

Ex: 1 3 2 4 5

sorted is 1 2 3 4 5

1 and 4 are in the correct place , whereas 3 and 2 are not in the correct place.

So the result would be

2, 3

Since 2 are incorrect and 3 are correct

Q2) Had to find the number of golden number in a given range where the golden number meant the

sum of the digits of a number - max number = max number

Ex: 325 is a golden number

$(3+2+5) - 5 = 5$

2. **Technical Round 1** (45 candidates)

The interview started with the interviewer asking me about my family background and where I originally come from. He then asked me for my introduction, which I had prepared in advance.

“My name is Maaz Malik, and I am currently in my 7th semester at this institute.

[mention of patents]

[mention of achievements—projects and hackathons]

[mention of hobbies]”

1. The interviewer then asked me to explain how RAG (Retrieval-Augmented Generation) works, as it was mentioned on my resume. I had worked on RAG as part of my internship.
2. He asked me what I was good at. I mentioned Java as my primary language.
3. Implementation of an in-memory key-value cache where each element has a TTL (time to live), and once the TTL expires, the element is automatically evicted. I provided a Java implementation for the same.

```

import java.util.*;

// Represents a cache item with a creation timestamp, TTL (time-to-live),
// and a key
class Item {
    long timestamp; // The time when the item was inserted
    int ttl;         // Time-to-live in seconds
    String key;      // The key associated with this item

    Item(String key, int ttl) {
        this.timestamp = java.time.Instant.now().getEpochSecond(); // Set
current time as creation time
        this.key = key;
        this.ttl = ttl;
    }
}

public class Main {

    // Map to store key-value pairs
    HashMap<String, Object> hmap = new HashMap<>();

    // Min-heap to track items based on TTL (soonest to expire first)
    PriorityQueue<Item> pq = new PriorityQueue<>((a, b) -> a.ttl - b.ttl);

    public static void main(String[] args) {
        Main mainObj = new Main();

        // Background thread to continuously check and remove expired
entries
        Thread t = new Thread(() -> {
            while (true) {
                mainObj.cooldown(200); // Sleep for 200 milliseconds
between checks

                synchronized (mainObj) { // Synchronize to avoid
concurrent modification issues
                    Item curr = mainObj.pq.peek(); // Peek at the
earliest-expiring item

```

```

        while (curr != null && mainObj.getSeconds() -
curr.timestamp >= curr.ttl) {
            // If the item has expired, remove it from both
map and queue

            mainObj.pq.poll();
            mainObj.hmap.remove(curr.key);
            System.out.println("removed key :" + curr.key);
            curr = mainObj.pq.peek(); // Check the next item
        }
    }
}

});

t.start(); // Start the cleanup thread

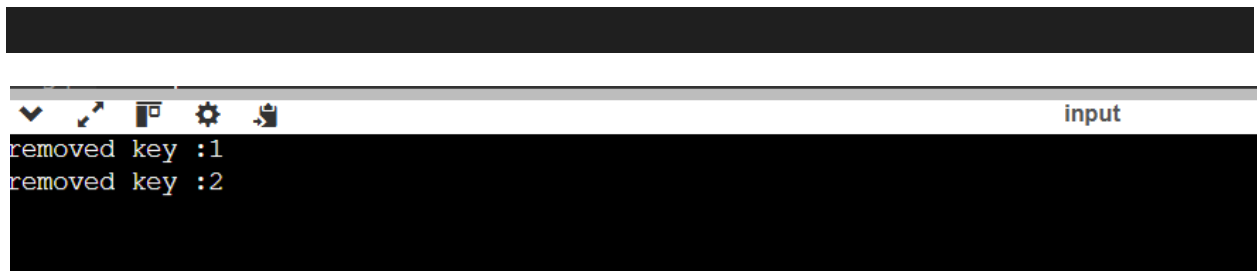
// Insert two items with different TTLs
mainObj.insert("1", "hello", 3); // expires after 3 seconds
mainObj.insert("2", "world", 5); // expires after 5 seconds
}

// Inserts a key-value pair with TTL into the cache
public synchronized void insert(String key, Object o, int ttl) {
    hmap.put(key, o);
    pq.add(new Item(key, ttl));
}

// Pauses the thread for the given duration
public void cooldown(int millis) {
    try {
        Thread.sleep(millis);
    } catch (Exception e) {
        // ignore
    }
}

// Returns current time in seconds since epoch
private long getSeconds() {
    return java.time.Instant.now().getEpochSecond();
}
}

```



```
removed key :1
removed key :2
```

4. How do lambda expressions work in Java? I was unable to answer this question.
5. Internal implementation of a PriorityQueue.
6. What are the four pillars of object-oriented programming, and what is abstraction?
7. Why do we extend abstract classes, and why do we implement interfaces? The answer was later provided by the interviewer: interfaces are contracts—you can implement a contract, but extending a contract does not make sense.
8. I had carried my patent along with my resume. I presented it to the interviewer, based on which he asked me some questions.

Always ask questions at the end of an interview. I asked him why there wasn't any mention of AI in the job description, since I am from the CSE AIML branch.

If the interviewer does not introduce themselves, always ask for their name, as it might be asked in the next round.

2. **Technical Round 2** (22 candidates)

The second technical round began with the interviewer asking me to introduce myself.

1. I was asked to code the **right view of a binary tree** on their laptop.
2. The interviewer then asked me to solve the same problem **without using a Queue**, suggesting a **DFS approach**.

```
private void dfs(TreeNode node, int depth, List<Integer> result) {  
  
    if (node == null) return;  
  
    if (depth == result.size()) {
```

```

        result.add(node.val);

    } else {

        result.set(depth, node.val);

    }

    dfs(node.left, depth + 1, result);

    dfs(node.right, depth + 1, result);

}

```

3. I was asked to write a function to verify whether a given tree is a BST(binary search tree). All elements to the left must be smaller than the root and all elements to the right must be greater than the root.

```

private boolean validate(TreeNode node, long min, long max) {

    if (node == null) return true;

    if (node.val <= min || node.val >= max) return false;

    return validate(node.left, min, node.val) &&
validate(node.right, node.val, max);

}

```

4. Next, I was given a **SQL question** involving two tables: **employees** and **customers**, each having **id**, **name**, **salary**, and **department**. Since departments could be common across both tables, the task was to **find the employee name and salary, with the maximum salary from each department** from both tables.

```

CREATE VIEW unified_people AS

SELECT id, name, salary, department FROM employees

UNION ALL

```

```
SELECT id, name, salary, department FROM customers;
```

```
SELECT name, salary, department
FROM unified_people up
WHERE salary = (
    SELECT MAX(salary)
    FROM unified_people
    WHERE department = up.department
);
```

5. The discussion then moved to **distributed computing**, specifically the **master-slave architecture**, based on one of the projects mentioned on my resume.
6. I was also asked about my **patents**.
7. I raised a question about the **absence of AI** in the job description, considering I come from the **CSE AIML branch**.

Tip: Always remember to ask for the interviewer's name if they don't introduce themselves, as it might be asked in a subsequent round.

3. **HR round** (9 candidates)

The interview began with a standard self-introduction prompt, followed by the question, *"Why should we hire you?"* To prepare for this, I researched the company's values and culture using Perplexity Pro a day in advance, which helped me tailor my response to align with their mission and expectations.

“If your manager is extremely busy and unable to assign you a project, how would you proceed?”

“You’ve invested significant time and effort into a project, but your manager decides to scrap it and reassign you. How do you respond?”

4. Technical Round 3

This round took place online around 8.45 PM from home. It was a short round of around 10 minutes. It started with the interviewer asking me about myself. As always I responded with the introduction I had prepared beforehand.

1.What is stack and heap memory.

2.Garbage collection in Java (different types of memories- old gen, new gen, permgen, metaspace)

3.What is clustering

RESOURCES

I spent a week preparing for aptitude questions including quantitative, logical, verbal and non-verbal aptitude from the following sources:

Topic wise quantitative aptitude

<https://www.indiabix.com/aptitude/questions-and-answers/>

<https://www.geeksforgeeks.org/aptitude/aptitude-questions-and-answers/>

Aptitude time tests(refer for data interpretation problems and for practice)

<https://www.indiabix.com/online-test/categories/>

Aptitude preparation (youtube)(prioritize career ride)

<https://www.youtube.com/watch?v=hlyal4sR0m8&list=PL8p2I9GklV454LdGfDOw0KkNazKuA-6B2>

<https://www.youtube.com/watch?v=tnc9ojITRg4&list=PLpyc33gOcbVA4qXMoQ5vmhefTruk5t9lt>

JAVA

https://www.youtube.com/watch?v=BSVKUk58K6U&list=PL9gnSGHSqcno1G3XjUbWzXHL8_EttOuKk

Common interview questions in JAVA

<https://drive.google.com/drive/folders/1pB0VVklXksdpDpRxzSdvWzJwC1BNM-4k?usp=sharing>

DBMS

<https://www.youtube.com/watch?v=dl00fOOYLOM&t=248s>

DSA

<https://takeuforward.org/strivers-a2z-dsa-course/strivers-a2z-dsa-course-sheet-2/>

SQL

<https://leetcode.com/studyplan/top-sql-50/>

<https://www.geeksforgeeks.org/sql/sql-query-interview-questions/>

<https://www.geeksforgeeks.org/sql/sql-interview-questions/>

In case of any further doubts, you can refer to the past interview experiences shared by TPO.